Operation Ember Ascent: Two-Stage Sounding Rocket for Mustafar

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Body material: PVC

1. Summary

This report explains a two-stage sounding rocket that was designed in OpenRocket to carry a 1 kg payload to around 4000 m on Mustafar. The final design reaches an apogee of 4077 m. Each section lands at 7.13 m/s under a parachute.

2. Design approach

• Body material: We have used PVC as our body material as it is light weighted, strong, easily accessible, cheap, smooth surface

• Geometry: length 1917 mm ,max.diameter 90.7 mm, nose cone type Ogive, nose length 378 mm, 4 fins per stage.

• Staging: booster with K76WN-P-0 and sustainer with 1190J355-17A-7. Total impulse 2539 N·s .

• Payload: 1.00 kg mass component in the payload bay no manual mass overrides used.

3. Thought process

We started the work by first making the model of the rocket in the openrocket in normal earth environment. Build two stages, adding the nose cone, two body tube, adjusting the diameters, then inner tubes with 4 centered rings , and when the base model was ready then we add then we add the motors of the stages added the payloads and then parachutes to land all the things safely. We tested it in the normal earth condition. After the successful simulation, we change the environment to the environment of the planet “mustafar”. As we change the condition, the rocket simulation showed the error as it was not taking the flight. So we changed the motors and adjusted the fins length of the body for the simulation to work. We then change the diameter of the parachute to make the payload and the second stage to land at the speed between 4 to 8 m-s . then adjusted the body and fins again to set apogee around 4000 m

4. Mustafar launch. Recovery system

Launch conditions

Parameter	Value	How it was set
Gravity	6.50 m/s²	Planetary Environment extension (custom planet JSON)
Pressure	75 kPa	Launch conditions
Temperature	700 K	Launch conditions
Density	0.374 kg/m³	Follows from pressure and temperature
Wind	10 m/s at 270°	Launch conditions
Launch altitude	0 m	Launch conditions

Recovery system

Section	Parachute	Diameter	Deployment	Ground speed
Payload	Parachute given	329 mm	Apogee	7.13
Sustainer	Parachute given	329 mm	Apogee	7.13
Booster	No parachute	NA	NA	NA

5. Results versus requirements

Requirement	Target	Result	Met?
Apogee	3900 to 4100 m	4077 m	Yes
Payload mass	0.95 to 1.05 kg	1.035 kg	Yes
Length	≤ 2200 mm	1917 mm	Yes
Total impulse	≤ 2800 N·s	2539 N-s	Yes
Deployment at apogee	Yes	Yes	Yes
Descent velocity	4 to 8 m/s	7.13 m/s	Yes
Total mass	≤ 11500 g	8077 g	Yes

6. Earth, versus Mustafar

Run	Apogee	Max velocity	Stability	Descent rate
Earth (all standard)	Around 6000 m	Around 150 m/s	1.81	Around 20 m/s
Mustafar	4077	253 m/s	.873	7.13 m/s

• Gravity (6.5 vs 9.81): weight is lower so the rocket climbs higher. With the burnout speed the drag-free height equals v squared divided by 2 times Gravity, which is about one and a half times greater.

• Density (0.374 vs 1.225 about ×0.31): drag is less so the rocket goes faster and reaches an apogee. However a parachute must have a larger area because terminal speed depends on the square root of (m times g divided by Density). For the parachute the terminal speed is about one and a half times faster.

• Temperature (700 K vs 288 K): the speed of increases from roughly 340 meters per second to about 530 meters per second. Because of this the same rocket speed corresponds to a Mach number.

• Pressure mainly changes density so keep Pressure from the Temperature test as OpenRocket permits.

• Wind (10 meters per second): weathercocking tilts the rocket into the wind causing drift. A stage that is barely stable may tumble.

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